

CHAPTER-1

ASSESSMENT TOOL -3

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SIMATS Online Programming Lab
DS PROGRAMMING

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C Programming Output Prediction

QUESTIONS

Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Q8
Second Largest Element
1 mark

Q9

Q1 Reverse a list 1 mark

Write a C program that takes an array as input and returns it reversed.

Your Code

```

1 #include <stdio.h>
2
3 int main() {
4     int n, i, temp;
5     scanf("%d", &n);
6     int a[n];
7     for(i = 0; i < n; i++) {
8         scanf("%d", &a[i]);
9     }
10    for(i = 0; i < n / 2; i++) {
11        temp = a[i];
12        a[i] = a[n - 1 - i];
13        a[n - 1 - i] = temp;
14    }
15    for(i = 0; i < n; i++) {
16        printf("%d ", a[i]);
17    }
18    return 0;
19 }
20

```

Submitted

Input

3
1
2
3

Output

3 2 1

Q2)

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C Programming Output Prediction

QUESTIONS

Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Q8
Second Largest Element
1 mark

Q9

Q2 Reverse a string 1 mark

Write a C program for palindrom

Your Code

```

1 #include<stdio.h>
2 #include<string.h>
3 int main(){
4     char str[100],temp;
5     int i=0,length=0;
6     scanf("%s",str);
7     while(str[length]!='\0')
8         length++;
9     for(i=0;i<length/2;i++){
10        temp=str[i];
11        str[i]=str[length-1-i];
12        str[length-1-i]=temp;
13    }
14    printf("%s",str);
15 }

```

Submitted

Input

length

Output

Q1)



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Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Q8
Second Largest Element
1 mark

Q9
Occurrence of an element
1 mark

Q10
Missing Element

Write a C program for Searching an Element in a Sorted Array

Your Code

```

1 #include<stdio.h>
2 int main() {
3     int arr[100], n, key, low, high, mid, found=0;
4     scanf("%d", &n);
5     for(int i=0; i<n; i++) {
6         scanf("%d", &arr[i]);
7     }
8     scanf("%d", &key);
9     low=0;
10    high=n-1;
11    while (low<=high) {
12        mid=(low+high)/2;
13        if(arr[mid]==key){
14            printf("element found at index %d\n", mid);
15            found=1;
16            break;
17        }
18        else if (key<arr[mid]){
19            high=mid-1;
20        }
21        else{
22            low=mid+1;
23        }
24    }
25    if(found==0){
26        printf("element not found");
27    }
28 }
```

Submitted

Input

```

5
1
2
3
.
```

Output

```

element found at index 3
```

Q3)

Q4)



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Q1
Reverse a list
1 mark

Q2
Reverse a string
1 mark

Q3
Searching
1 mark

Q4
Searching
1 mark

Q5
Array
1 mark

Q6
Merging
1 mark

Q7
Largest and Smallest element
1 mark

Q8
Second Largest Element
1 mark

Q9
Occurrence of an element
1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```

1 #include<stdio.h>
2 int main() {
3     int n, a[100], count=0, i, x;
4     scanf("%d", &n);
5     for(i=0; i<n; i++) {
6         scanf("%d", &a[i]);
7     }
8     scanf("%d", &x);
9     for (i=0; i<n; i++) {
10        if (x==a[i]) {
11            count++;
12        }
13    }
14    printf("the occurrence of the element is %d", count);
15 }
```

Submitted

Input

```

10
1
2
3
.
```

Output



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Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Q8

Second Largest Element

1 mark

✓

Q9

Occurrence of an element

1 mark

✓

Q10

Missing Element

1 mark

✓

10/10 answered

Your Code

```

1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key, low, high, mid, i;
5     scanf("%d", &n);
6
7     for(i = 0; i < n; i++) {
8         scanf("%d", &a[i]);
9     }
10
11     scanf("%d", &key);
12
13     low = 0;
14     high = n - 1;
15
16     while(low <= high) {
17         mid = (low + high) / 2;
18
19         if(a[mid] == key) {
20             printf("Element found at index %d", mid);
21             return 0;
22         }
23         else if(a[mid] < key)
24             low = mid + 1;
25         else
26             high = mid - 1;
27     }
28     printf("Element not found");
29
30     return 0;
31 }

```

Submitted

Input

5
1
2
3

Output

Q5)

Q6)



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QUESTIONS

Q1

Reverse a list

1 mark

✓

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Q8

Second Largest Element

1 mark

✓

Q9

Occurrence of an element

1 mark

✓

Q6 Merging

1 mark

Write a C program for Merge Two Arrays

Your Code

```

1 #include<stdio.h>
2 int main(){
3     int a[100],i,n1,b[100],n2,c[100];
4     scanf("%d",&n1);
5     for(i=0;i<n1;i++){
6         scanf("%d",&a[i]);
7     }
8     scanf("%d",&n2);
9     for(i=0;i<n2;i++){
10        scanf("%d",&b[i]);
11    }
12    for(i=0;i<n1;i++){
13        c[i]=a[i];
14    }
15    for(i=0;i<n2;i++){
16        c[i+n1]=b[i];
17    }
18    for(i=0;i<n1+n2;i++){
19        printf("%d ",c[i]);
20    }
21 }

```

Submitted

Input

3
10
20
30

Output



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QUESTIONS

Q1

Reverse a list

1 mark

✓

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Q8

Second Largest Element

1 mark

✓

Q9

Occurrence of an element

1 mark

✓

Q7

Largest and Smallest element

1 mark

Write a C program to Find the Largest and Smallest Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main(){
3     int i,n,a[100],j,temp;
4     scanf("%d",&n);
5     for(i=0;i<n;i++){
6         scanf("%d",&a[i]);
7     }
8     for(i=0;i<n-1;i++){
9         for(j=0;j<n-1-i;j++){
10
11             if(a[j]>a[j+1]){
12                 temp=a[j];
13                 a[j]=a[j+1];
14                 a[j+1]=temp;
15             }
16         }
17     }
18     printf("the largest number %d\n",a[n-1]);
19     printf("the smallest number is %d",a[0]);
20
21 }
```

Input

5
23
67
45
...

Output

Submitted

Q8)



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QUESTIONS

Q1

Reverse a list

1 mark

✓

Q2

Reverse a string

1 mark

✓

Q3

Searching

1 mark

✓

Q4

Searching

1 mark

✓

Q5

Array

1 mark

✓

Q6

Merging

1 mark

✓

Q7

Largest and Smallest element

1 mark

✓

Q8

Second Largest Element

1 mark

✓

Q9

Occurrence of an element

1 mark

✓

Q8

Second Largest Element

1 mark

Write a C program to Find the Second Largest Element

Your Code

```
1 #include<stdio.h>
2 int main(){
3     int n,i,j,temp,a[100];
4     scanf("%d",&n);
5     for(i=0;i<n;i++){
6         scanf("%d",&a[i]);
7     }
8     for(i=0;i<n-1;i++){
9         for(j=0;j<n-1-i;j++){
10
11             if(a[j]>a[j+1]){
12                 temp=a[j];
13                 a[j]=a[j+1];
14                 a[j+1]=temp;
15             }
16         }
17     }
18
19     printf("the second largest number is %d",a[n-2]);
20
21 }
```

Input

5
3
4
2
...

Output

Submitted

Q7)

C Programming Output Prediction

← Back

QUESTIONS

Q1
Reverse a list
1 markQ2
Reverse a string
1 markQ3
Searching
1 markQ4
Searching
1 markQ5
Array
1 markQ6
Merging
1 markQ7
Largest and Smallest element
1 markQ8
Second Largest Element
1 mark

Q9 Occurrence of an elemnet

1 mark

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[100], n, x, i, first=-1, last=-1;
4     scanf("%d", &n);
5     for(i=0; i<n; i++) {
6         scanf("%d", &a[i]);
7     }
8     scanf("%d", &x);
9     for(i=0; i<n; i++) {
10        if(a[i]==x) {
11            if(first==-1)
12                first=i;
13            last=i;
14        }
15    }
16    if(first==-1)
17        printf("not found");
18    else
19        printf("first=%d\nlast=%d", first, last);
20 }
21 }
```

Submitted

Input

Output

Q9)

Q10

C Programming Output Prediction

← Back

QUESTIONS

Q1
Reverse a list
1 markQ2
Reverse a string
1 markQ3
Searching
1 markQ4
Searching
1 markQ5
Array
1 markQ6
Merging
1 markQ7
Largest and Smallest element
1 markQ8
Second Largest Element
1 mark

Q10 Missing Element

1 mark

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[100], i, n;
4     scanf("%d", &n);
5     for(i=0; i<n-1; i++) {
6         scanf("%d", &a[i]);
7     }
8     for(i=0; i<n-2; i++) {
9         if(a[i+1] != a[i]+1) {
10            printf("the missing element is %d\n", a[i]+1);
11        }
12    }
13 }
```

Submitted

Input

Output